

Software Requirements Specification (SRS)

‘AI Crop Disease Detection and Treatment System’

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1. Introduction

Software Requirement Specification (SRS) is a document that describes the functional and non-functional requirements of a software system. It provides a clear understanding of how the system should work and what features it must include.

The proposed system **AI Crop Disease Detection and Treatment Recommendation System** helps farmers detect crop diseases using image processing and artificial intelligence. The user can upload an image of a crop leaf, and the system will analyze the image to identify the disease and suggest possible treatments.

This system aims to help farmers identify diseases early and improve crop productivity.

2. Overall Description

The system is designed to detect plant diseases using **machine learning and image processing techniques**.

Users upload an image of a plant leaf (such as tomato, spinach, ladyfinger, brinjal, or arecanut). The system processes the image and predicts whether the leaf is healthy or diseased. If a disease is detected, the system provides information about the disease and recommended treatment methods.

Main Users

- Farmers
- Agricultural researchers
- Students and researchers in agriculture

2.1 Product Perspective

The system will be a **web-based or desktop application** that integrates:

- Image upload module
- Machine learning model
- Disease detection system
- Treatment recommendation module

2.3 Product Functions

- Upload leaf image
- Detect crop disease
- Display disease name
- Recommend treatment
- Store results in database

2.3 User Characteristics

Users may have **basic computer knowledge** but may not be experts in technology.

3. Special Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4GB RAM
- Camera or image input
- Internet connection

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python
- Libraries: TensorFlow / Keras / OpenCV
- Development Tools: Jupyter Notebook / VS Code
- Database: MySQL / SQLite

3.1 User Interfaces

The **AI Crop Disease Detection and Treatment Recommendation System** provides a simple and user-friendly interface for users. Farmers or users can upload images of crop leaves through the system interface. The system displays the detected disease and recommended treatment clearly on the screen. The interface includes basic features such as **image upload option, result display section, and navigation buttons**. It is designed to be easy to use so that users can quickly analyze crop diseases and view the results.

3.2 Hardware Interfaces

The **AI Crop Disease Detection and Treatment Recommendation System** interacts with basic hardware devices such as computers, smartphones, and cameras. These devices are used to capture and upload images of crop leaves to the system.

The system processes the images using the computer's processor and memory. An internet connection may be used to access the web application and transfer image data for analysis.

3.3 Software Interfaces

The **AI Crop Disease Detection and Treatment Recommendation System** interacts with several software components to perform its functions.

The system uses **machine learning libraries** such as Python-based frameworks to analyze crop leaf images and detect diseases. It also connects to a **database system** to store crop images, disease information, and treatment recommendations. The application runs on an **operating system such as Windows or Linux** and uses development tools like Python and supporting libraries.

Data exchanged between components includes **uploaded crop images, processed image data, detected disease information, and treatment recommendations**. These components communicate within the system to ensure accurate disease detection and result generation.

4. System Features

4.1 Image Upload Feature

Description:

- This feature allows the user to upload an image of a crop leaf to the system for disease detection.

Inputs:

- Leaf image (JPG / PNG)

Processing:

- The system reads and preprocesses the uploaded image.

Outputs:

- The image is sent to the disease detection model.

4.2 Disease Detection Feature

Description:

- This feature analyzes the uploaded crop image using a trained machine learning model to detect whether the plant is healthy or affected by a disease.

Inputs:

- Preprocessed crop leaf image.

Processing:

- The machine learning model analyzes patterns in the image.
- The system compares the image with the trained dataset.

Outputs:

- Disease name or Healthy result.

4.3 Treatment Recommendation Feature

Description:

- After detecting the disease, the system provides recommended treatments to control or cure the disease.

Inputs:

- Detected disease name.

Processing:

- The system retrieves treatment information from the database.

Outputs:

- Recommended pesticides or treatment methods.

4.4 Crop Information Display

Description:

- The system provides information about the detected crop disease.

Inputs:

- Disease identification result.

Processing:

- Retrieve disease details from the database.

Outputs:

- Disease description
- Causes of disease
- Prevention methods

4.5 Result Display Feature

Description:

This feature displays the final result to the user.

Outputs:

- Uploaded leaf image
- Disease prediction
- Treatment suggestions

4.6 Dataset Management (Admin Feature)

Description:

The admin can manage crop datasets used for training the AI model.

Functions:

- Add new crop images
- Update disease data
- Improve model accuracy

5. Other Non-Functional Requirements

5.1 Performance Requirements

- The system should process and detect disease within **5–10 seconds** after image upload.
- The system should support multiple image inputs.
- The prediction accuracy should be as high as possible.

5.2 Safety Requirements

- The system should ensure that user data and images are safely stored.
- Incorrect predictions should not cause system failure.
- The system must prevent loss of data.

5.3 Software Quality Attributes

Reliability

The system should provide consistent and accurate disease detection results.

Usability

The system should be easy to use for farmers and students.

Maintainability

The system should allow easy updates for adding new crop diseases.

Portability

The system should run on different operating systems such as Windows and Linux.

5.4 Security Requirements

- Only authorized users should access the system.
- User data and images must be protected.
- The system should prevent unauthorized data modification.
- Users must upload **clear images of crop leaves**.
- The system only supports **specific crops included in the dataset**.
- Disease detection results depend on the trained AI model.

5.5 Business Rules

6. Other Requirements

Database Requirements

The system must maintain a database containing:

- Crop names
- Disease names
- Disease descriptions
- Treatment recommendations
- Uploaded images and results

Future Enhancements

- Mobile application support
- More crop disease detection
- Real-time detection using mobile camera
- Integration with agricultural advisory systems